ANP v6.3 Jet Release Holdout

A source-separated check of PNMF's production Extra Trees and Random Forest models. Models train only on legacy v2.3 Jet curves after a predeclared family purge and are scored on three frozen v6.3 reference curves.

76 / 3	1,160	3.953	3.900
legacy train / v6.3 test curves	scored power-distance cells	ET pooled RMSE, dB	RF pooled RMSE, dB

Official-source conclusion

v6.3 is preferable as newer EASA-collected/verified reference provenance, but no official source proves universal accuracy superiority.

This is a provenance and chronology judgment, not a universal curve-accuracy claim. Both model scores are close in this small holdout; the category tables are more informative than a winner label.

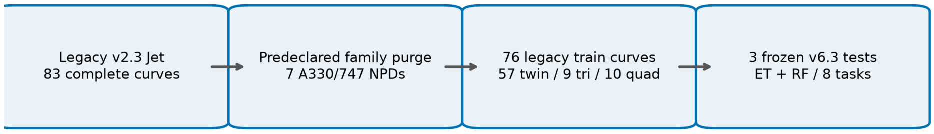
Evidence boundary

- Only three independent curves are tested; the 1,160 cells are correlated.
- Tri- and quad-engine categories each have one v6.3 candidate only.
- Use for conceptual screening and model scrutiny, never certification.

Sources, Selection, and Split

EASA states that it collects, verifies and makes ANP information available under Regulation (EU) No 598/2014. EASA separately labels v2.3 as legacy data assembled before that mandate.

EASA Aircraft Noise and Performance (ANP) | EASA ANP legacy data | Regulation (EU) 598/2014



v6.3 targets are absent from training and descriptor selection never uses noise levels; the physics route remains independent.

Eng.	Frozen NPD / ACFT_ID	Description	Distance
2	A330-743L / A330-743L	Airbus A330-743L / RR Trent 772B	0.216692
3	FAL900EX / FAL900EX	Dassault FAL900EX / TFE731-60	0.000000
4	747400RN / 747400RN	Boeing 747400RN / PW4062A	0.000000

Descriptor-only rule: per-category IQR-scaled Euclidean distance over log10(MTOW), log10(total static thrust), and noise chapter. Zero-IQR terms contribute zero; lexical NPD_ID breaks ties. Targets and errors never select a reference.

Evidence unit	Train	Test	Interpretation
Jet curves	76	3	v2.3 only / v6.3 only
Approach rows per metric	216	12	Power rows
Departure rows per metric	297	17	Power rows
All test cells	-	1,160	116 rows x 10 distances

Separation and Learning Method

Predeclared family guard

Before fitting, seven legacy NPDs are excluded: **CF680E, TRENT7, JT9DBD, JT9DFL, JT9D7Q, PW4056, and GENX67**. They conservatively guard the A330 and 747 families represented in the v6.3 holdout. Falcon 20 is not automatically purged; no broad name heuristic is used.

Stage	Twin	Tri	Quad	Total
Legacy complete-task Jet pool	59	9	15	83
Predeclared exclusions	2	0	5	7
Training after purge	57	9	10	76
Frozen v6.3 test	1	1	1	3

Models and tasks

Route	Configuration	Coverage
Extra Trees	500 trees; depth 24; features 0.5; leaf 1	8 tasks
Random Forest	200 bootstrap trees; leaf 2	8 tasks

Each task predicts ten standard NPD distances. Inputs retain the frozen 12-feature learned surface and mixed-power-unit correction. Predictions retain the non-increasing distance projection.

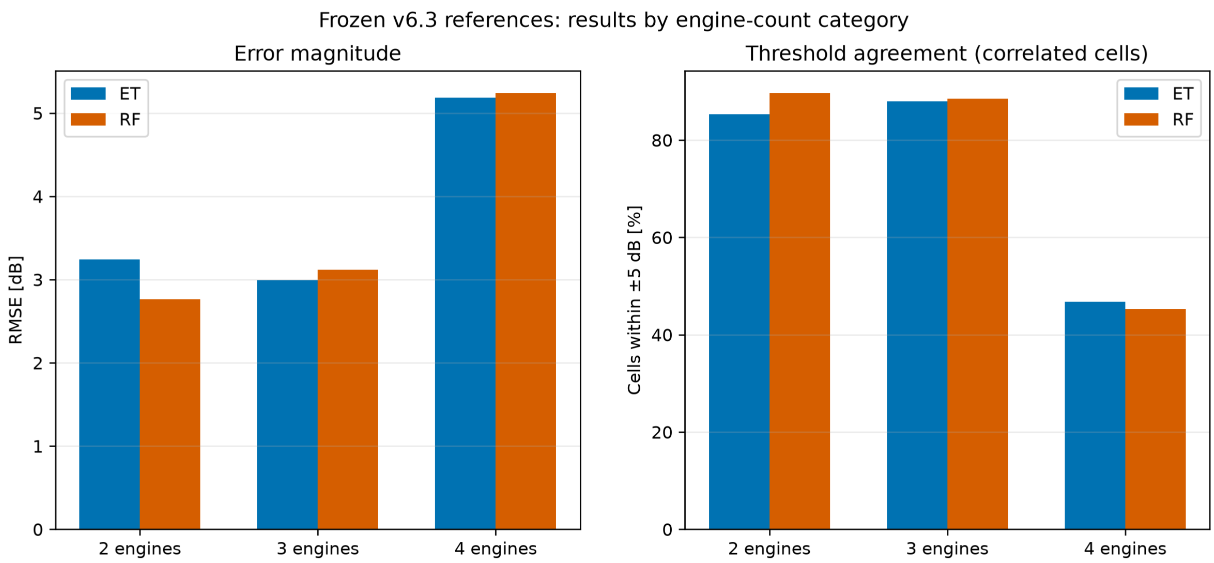
Scoring formulas

```
error = prediction - truth
RMSE = sqrt(mean(error^2)); MAE = mean(abs(error))
bias = mean(error); p90 = percentile(abs(error), 90)
within +/-k dB = 100 x mean(abs(error) <= k)
```

PhysicsNPDModel remains an independent SEL/LAmax component-source workflow. It supplies neither features nor targets to ET/RF here.

Overall Measured Results

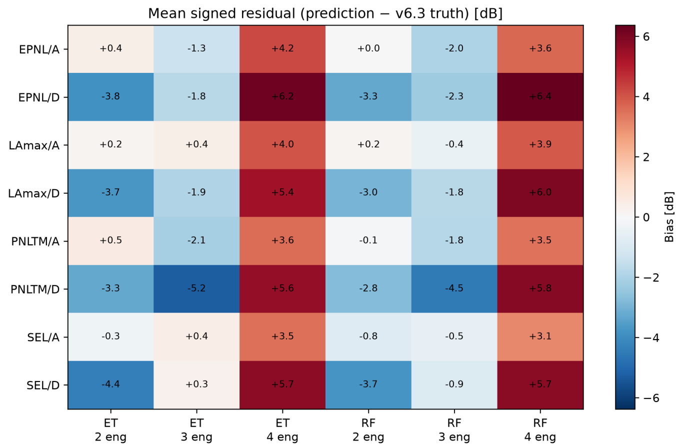
Model	Aggregation	RMSE	MAE	Bias	P90 abs	+/-3	+/-5
ET	Cell pooled	3.953	3.239	+0.545	6.425	50.000%	72.931%
ET	Category balanced	3.931	3.213	+0.459	6.010	50.648%	73.343%
RF	Cell pooled	3.900	3.231	+0.491	6.398	51.121%	73.966%
RF	Category balanced	3.868	3.193	+0.413	5.785	51.972%	74.491%

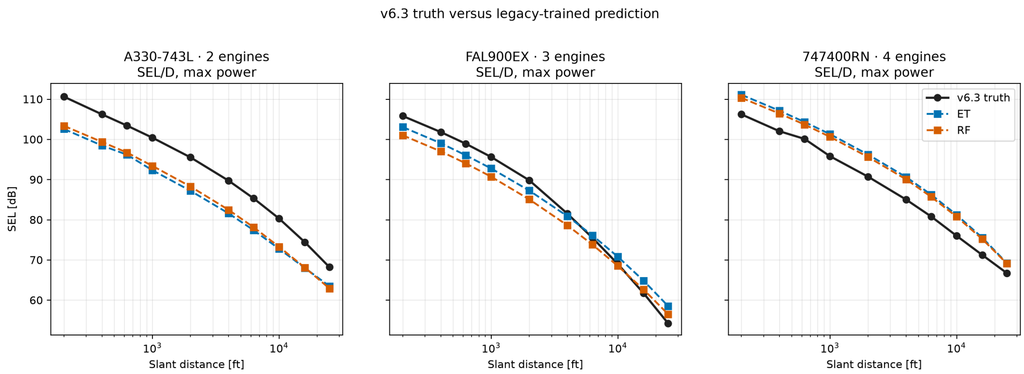


Cell-pooled metrics weight every cell equally. Category-balanced metrics give twin, tri, and quad categories equal influence; balanced RMSE is the square root of mean category MSE. "Within +/-3 dB" and "within +/-5 dB" are threshold-agreement percentages over correlated cells, not success probabilities or certification margins.

Reference-Level Error Patterns

Model	Eng.	RMSE	MAE	Bias	P90	+/-3	+/-5
ET	2	3.240	2.440	-2.021	5.972	69.44%	85.28%
ET	3	2.993	2.232	-1.562	5.386	74.50%	88.00%
ET	4	5.187	4.966	+4.961	6.672	8.00%	46.75%
RF	2	2.765	2.087	-1.855	5.222	76.67%	89.72%
RF	3	3.121	2.503	-1.894	5.271	67.00%	88.50%
RF	4	5.244	4.987	+4.987	6.863	12.25%	45.25%





The lower chart is an illustrative maximum-power SEL departure slice. The four-engine reference shows a positive bias for both learners. All eight tasks and every power row remain in predictions.csv.

Validity Boundaries and Audit Trail

v6.3 is preferable as newer EASA-collected/verified reference provenance, but no official source proves universal accuracy superiority.

Scientific limitations

- Three independent curves are too few for fleet-wide or certification claims.
- The 1,160 cells are repeated observations within those curves and are correlated.
- Tri- and quad-engine categories each have one v6.3 candidate; neither is a general representative.
- Selection uses only weight, installed thrust, and noise chapter.
- The family purge reduces obvious A330/747 leakage but cannot prove absence of all engineering similarity.
- Results do not establish performance for unconventional aircraft or propulsion.

Reproducible evidence

Seed 20260724. Datastore SHA-256 `9b3ea2f58347ebb348128c49b3238e6a0b28852858fa7e87c89fad51d5e9fe8e`. Candidate scores, reference metadata, exclusions, predictions, fit records, summaries, source manifest, official URLs, and artifact hashes are saved under `outputs/model_validation/jet_reference_v63`.

Decision boundary

Use this release holdout as a transparent diagnostic of legacy-trained ET/RF behavior on the three frozen v6.3 references. It does not replace broader grouped validation and must not be presented as universal accuracy or certification evidence.